Task-Aware Schatten Budgeting Across Resolution and Time in Latent Diffusion

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Paper under double-blind review

Abstract

Latent diffusion repeatedly applies Gaussian perturbations while representation resolution and diffusion time determine how many spectral directions can accumulate noise. A dimension-uniform Schatten-2 condition identifies the asymptotic feasible region, but it does not decide whether a finite model has already exceeded an acceptable risk level, nor which modes should receive a limited perturbation budget. We turn this threshold into a finite decision problem. For the whitened relative covariance $K_{R,t}$, raw $T_2 = \|K_{R,t}\|_{\mathcal{S}_2}^2$ gives the exact single-step log-density variance and, together with an operator-norm cap and confidence level, a finite tail certificate; maximum, path, and end-state budgets separate the distinct ways perturbations accumulate over a diffusion schedule. We then propose *task-aware Schatten budgeting*: allocate a prescribed T_2 budget by minimizing task-weighted spectral distortion under positivity and operator-norm constraints. The resulting shrinkage departs least from the target perturbation on modes with high task sensitivity—including task-critical high frequencies—while controlling cumulative covariance mass rather than frequency alone. A structured latent gate implements the allocation, with exact accounting for structured covariances and cost-accounted stochastic audits for implicit ones. This framework turns the theorem's feasible region into an actionable design constraint and makes quality–stability tradeoffs explicit operating choices for latent diffusion.

1 Introduction

Gaussian noise is usually exposed as a scalar control knob. In latent diffusion, however, that scalar acts on every resolved latent direction and is applied repeatedly over time. Increasing image resolution or the number of diffusion steps can therefore change the aggregate perturbation even when the nominal noise scale is fixed. The practical question is not simply whether the noise is "small," but how much covariance risk is acceptable and which latent modes should receive that budget.

We make this accumulation explicit through relative covariance geometry. For a reference covariance C_R and a perturbed covariance $C_R + \Delta C_R$ at resolution R, define

$$K_R = C_R^{-1/2} \Delta C_R C_R^{-1/2}, \qquad T_2(R) := \|K_R\|_{\mathcal{S}_2}^2 . \tag{1}$$

The eigenvalue contribution to Gaussian covariance KL is $\lambda - \log(1+\lambda)$, so its small-mode tail is quadratic. Under uniform positivity and operator-norm bounds, we show that $\sup_R T_2(R) < \infty$ is exactly the no-divergence region for this objective and that the centered log-density ratio has variance $T_2(R)/2$. This is a dimension-uniform statement: every fixed matrix is Schatten, but a refinement sequence need not have a uniform Schatten budget.

That asymptotic boundary is not yet a deployment rule. A finite system needs a numerical risk certificate, and a diffusion schedule creates several different notions of accumulation. We therefore pair raw T_2 with an operator-norm cap, a positivity margin, and a confidence level, and distinguish

$$\mathcal{B}_{\max} = \max_{1 \le t \le T} \mathbb{E} T_2(K_{R,t}), \quad \mathcal{B}_{\text{path}} = \sum_{t=1}^{T} \mathbb{E} T_2(K_{R,t}), \quad \mathcal{B}_{\text{end}} = T_2(K_{R,0\to T}). \tag{2}$$

They measure the worst transition, cumulative path risk, and final marginal distortion, respectively. A timestep-weighted training diagnostic is reported separately because it is generally not path KL.

The remaining problem is allocation. Uniform shrinkage suppresses useful and redundant modes alike; a frequency cutoff can obtain stability by deleting fine detail; an operator-norm cap leaves growing multiplicity unchecked. We instead propose *task-aware Schatten budgeting*:

$$\widetilde{K}_\tau = \underset{\widetilde{K}\succeq 0}{\arg\min}\, D_{\mathrm{task}}(\widetilde{K}, K) \quad \text{s.t.} \quad \left\|\widetilde{K}\right\|_{\mathcal{S}_2}^2 \le \tau, \quad \left\|\widetilde{K}\right\|_{\mathrm{op}} \le \rho_{\mathrm{cap}}. \tag{3}$$

In a fixed structured basis with nonnegative task sensitivities q_j, the KKT solution is $\widetilde{\lambda}_j = \min\{\rho_{\mathrm{cap}}, q_j\lambda_j/(q_j+\mu)\}$. It departs least from the target perturbation on task-sensitive modes, regardless of their physical frequency, and spends less budget on low-sensitivity multiplicity. Thus the method controls an operator ideal rather than assuming that low frequency is synonymous with importance.

We instantiate the allocation as a lightweight covariance gate in fixed whitened latent coordinates. Structured gates are audited exactly; implicit encoder and diffusion operators use an independent, cost-accounted probe bank. The empirical design has two jobs: a controlled high-frequency task separates the method from low-pass filtering, and one pinned latent-diffusion system tests the encoder and the diffusion path over a common resolution–time grid.

Our contributions are:

- **A finite resolution–time certificate.** We turn the sharp Schatten-2 boundary into raw, shell-wise, maximum, path, and endpoint quantities that remain comparable as latent resolution and diffusion time change.
- **Task-aware Schatten budgeting.** We derive a closed-form allocation rule under squared Schatten and operator-norm constraints that shrinks high-sensitivity modes less regardless of frequency. Its latent implementation combines exact structured accounting with independent, cost-accounted probes for implicit operators.

2 Gaussian Noise Under Refinement

2.1 Relative covariance and the finite certificate

At cutoff N, compare $p_N = \mathcal{N}(0, C_N)$ with $q_N = \mathcal{N}(0, C_N + \Delta C_N)$ and whiten by the positive reference:

$$K_N = C_N^{-1/2}\Delta C_N C_N^{-1/2}, \qquad I + K_N \succ 0. \tag{4}$$

The covariance contribution to Gaussian KL is

$$D_{\mathrm{KL}}(q_N \| p_N) = \frac{1}{2}\sum_{j\le n_N}\big[\lambda_{j,N} - \log(1+\lambda_{j,N})\big]. \tag{5}$$

Because the summand is quadratic near zero, define the operator functional and its value along the cutoff sequence by

$$T_2(K) := \|K\|_{\mathcal{S}_2}^2, \qquad T_2(N) := T_2(K_N) = \mathrm{Tr}(K_N^* K_N) = \sum_{j\le n_N} s_j(K_N)^2. \tag{6}$$

We report the finite certificate

$$\begin{aligned} \mathfrak{C}_N &= (T_2(N), M_N, \delta_N^{\mathrm{cert}}), \quad M_N = \|K_N\|_{\mathrm{op}}, \\ \delta_N^{\mathrm{act}} &= \lambda_{\min}(I+K_N) > 0, \quad \delta_N^{\mathrm{cert}} = \min\{1, \delta_N^{\mathrm{act}}\}. \end{aligned} \tag{7}$$

Raw T_2 sets the exact log-density variance scale, M_N controls the non-Gaussian tail, and δ_N^{cert} supplies the theorem's positivity constant; we also report the actual geometric margin δ_N^{act}. All are dimensionless after whitening. A confidence level completes the finite-risk statement in Theorem 3.1.

Every fixed matrix belongs to every finite-dimensional Schatten class; only a sequence can reveal refinement instability. For example, $C_N + \Delta C_N = aC_N$ gives

$$K_N = (a-1)I_{n_N}, \qquad T_2(N) = (a-1)^2 n_N, \tag{8}$$

so an update that is legal at each cutoff accumulates in every new direction. Likewise, $s_j(K) \asymp j^{-\alpha}$ has bounded squared mass only for $\alpha > 1/2$.

2.2 SHELL DIAGNOSTICS

A fitted decay exponent can hide alternating blocks, sparse spikes, or growing multiplicity. We therefore report dyadic rank-shell mass

$$\mathcal{B}^{\mathrm{rank}}_{N,\ell} := \sum_{2^\ell \le j < \min\{2^{\ell+1}, n_N+1\}} s_j(K_N)^2. \tag{9}$$

When a refinement-compatible physical decomposition exists, with $P_{N,\ell}$ projecting onto $2^\ell \le \|k\| < 2^{\ell+1}$, we separately use

$$\mathcal{B}^{\mathrm{phys}}_{N,\ell} := \|K_N P_{N,\ell}\|^2_{\mathcal{S}_2}, \qquad T_2(N) = \sum_\ell \mathcal{B}^{\mathrm{phys}}_{N,\ell}. \tag{10}$$

Rank is not spatial frequency: the first diagnostic is basis invariant, while the second asks where physical modes enter. Growth slopes are always labeled by their independent variable ($\log n_N$ or $\log R$).

If a continuum interpretation is claimed, the cutoffs must be compatible subspaces of one Hilbert space, the reference covariance must itself be positive, injective, and trace class, and the lifted updates must share a positivity gap. These conditions, the precision–covariance conversion, and the difference between boundedness and convergence are collected in Sections C and D. Finite neural-feature experiments use the diagnostics without asserting a white-noise continuum law.

3 CERTIFICATES AND TASK-AWARE BUDGETING

3.1 THE FINITE AND ASYMPTOTIC GUARANTEES

Write the covariance residue as

$$\mathcal{L}^{\mathrm{ren}}_N(K_N) := \frac{1}{2} \sum_{j \le n_N} [\lambda_{j,N} - \log(1 + \lambda_{j,N})]. \tag{11}$$

Its finite part is represented by $-\frac{1}{2}\log\det_2(I + K_N)$ (Section A). The following result turns $\mathfrak{C}_N = (T_2(N), M_N, \delta^{\mathrm{cert}}_N)$ into a decision certificate.

Theorem 3.1 (Finite raw-T_2 certificate)**.** *Let $K_N = K_N^*$, $I + K_N \succeq \delta I$ for $0 < \delta \le 1$, and $\|K_N\|_{\mathrm{op}} \le M$. Then*

$$\frac{T_2(N)}{4(1+M)^2} \le \mathcal{L}^{\mathrm{ren}}_N(K_N) \le \frac{T_2(N)}{4\delta^2}. \tag{12}$$

If $X \sim q_N$ and $Y_N = \log \frac{dq_N}{dp_N}(X) - \mathbb{E}_{q_N} \log \frac{dq_N}{dp_N}(X)$, then

$$\mathrm{Var}(Y_N) = \frac{1}{2} T_2(N), \qquad \Pr\Big(|Y_N| \ge \sqrt{T_2(N)x} + Mx\Big) \le 2e^{-x}. \tag{13}$$

For $K_N \succeq 0$, the objective constants sharpen to $[4(1+M)]^{-1}$ and $1/4$.

Thus T_2 is an exact variance budget but not, by itself, a complete tail certificate: M and a chosen failure probability control rare fluctuations, while δ controls proximity to the covariance boundary. Proofs and the explicit confidence-radius form are in Section B.

Theorem 3.2 (Uniform refinement threshold)**.** *For a refinement family with common (δ, M),*

$$\sup_N T_2(N) < \infty \iff \sup_N \mathcal{L}^{\mathrm{ren}}_N(K_N) < \infty \iff \sup_N \mathrm{Var}(Y_N) < \infty. \tag{14}$$

If $T_2(N) \to \infty$, the other two quantities diverge.

Uniform boundedness is a no-divergence statement, not convergence by itself. Compatible $\mathcal{S}_2$ convergence and the familiar $s_j \asymp j^{-\alpha}$ transition at $\alpha = 1/2$ are stated precisely in Section C.

3.2 ALLOCATE THE COVARIANCE BUDGET

Let B_N be a noise-amplitude operator in whitened coordinates. Its relative covariance is

$$K_N = B_N B_N^*, \qquad \Delta C_N = C_N^{1/2} K_N C_N^{1/2}. \tag{15}$$

Hence a diagonal amplitude b_j contributes b_j^4 to raw T_2. Throughout, τ denotes the *squared* budget $\|K_N\|_{\mathcal{S}_2}^2 \leq \tau$. The basis-invariant unweighted baseline is radial projection,

$$\Phi_\tau^{\mathrm{rad}}(K_N) = \gamma_N K_N, \qquad \gamma_N = \min\left\{1, \sqrt{\frac{\tau}{T_2(N)}}\right\}, \tag{16}$$

with the convention $\gamma_N = 1$ when $K_N = 0$. Our task-aware rule fixes a refinement-compatible basis, assumes the positive injected covariance is diagonal in that basis, and uses nonnegative entries $q_{j,N}$ from a local PSD task model (e.g., squared sensitivities or a Gauss–Newton diagonal):

$$\min_{0 \leq \widetilde{\lambda}_{j,N} \leq \rho_{\mathrm{cap}}} \frac{1}{2} \sum_j q_{j,N} (\widetilde{\lambda}_{j,N} - \lambda_{j,N})^2 \quad \text{s.t.} \quad \sum_j \widetilde{\lambda}_{j,N}^2 \leq \tau. \tag{17}$$

Its KKT solution is

$$\widetilde{\lambda}_{j,N} = \min\left\{\rho_{\mathrm{cap}}, \frac{q_{j,N}}{q_{j,N} + \mu} \lambda_{j,N}\right\}, \qquad \mu \geq 0, \tag{18}$$

where $\rho_{\mathrm{cap}} > 0$, bisection chooses μ when the budget is active, and the canonical value is zero if $q_{j,N} = \mu = 0$. Larger task weight means less departure from the target perturbation, independent of physical frequency. This is a closed-form commuting approximation, not a solution to general nonconvex task distortion; raw signed gradients are not valid weights. The derivation and modewise envelope variant are in Section E.

To implement the covariance solution with an amplitude factor, let $K_N = U_N \operatorname{diag}(\lambda_{j,N}) U_N^*$ and set

$$g_{j,N} = \sqrt{\frac{\widetilde{\lambda}_{j,N}}{\lambda_{j,N}}}, \qquad \widetilde{B}_N = U_N \operatorname{diag}(g_{j,N}) U_N^* B_N, \tag{19}$$

with $g_{j,N} = 1$ when both eigenvalues are zero. Then $\widetilde{B}_N \widetilde{B}_N^* = U_N \operatorname{diag}(\widetilde{\lambda}_{j,N}) U_N^*$; using covariance ratios directly as amplitude gains would square the intended shrinkage.

3.3 RESOLUTION AND DIFFUSION TIME

Let $K_{N,t}(X_{t-1})$ be the conditional relative covariance update between two Markov transitions, whitened by the conditional reference. We keep four quantities distinct:

$$\mathcal{B}_{\max}(N,T) := \max_{1 \leq t \leq T} \mathbb{E}_q T_2(K_{N,t}), \tag{20}$$

$$\mathcal{B}_{\mathrm{path}}(N,T) := \sum_{t=1}^{T} \mathbb{E}_q T_2(K_{N,t}), \tag{21}$$

$$\mathcal{B}_{\mathrm{train}}(N,T) := \sum_{t=1}^{T} w_t \mathbb{E}_{\pi_t} T_2(K_{N,t}), \tag{22}$$

$$\mathcal{B}_{\mathrm{end}}(N,T) := \left\| (C_{N,T}^p)^{-1/2} (C_{N,T}^q - C_{N,T}^p)(C_{N,T}^p)^{-1/2} \right\|_{\mathcal{S}_2}^2. \tag{23}$$

Under common certificate constants and $q_0 = p_0$, $\mathcal{B}_{\mathrm{path}}$ controls the conditional-transition covariance part of path KL and equals twice the variance of the centered covariance log-ratio martingale. An initial KL and conditional mean shifts, when present, are added separately. The weighted training quantity uses the estimator distribution π_t and is not path KL unless $w_t = 1$ and π_t is the path marginal. The endpoint follows the actual covariance recursion and is generally neither a sum nor a product of stepwise updates. Derivations and the weighted variance identity are in Section F.

3.4 Latent implementation and audit

For encoder Jacobian $J_N(x)$, input perturbation covariance Σ_N, and a fixed latent reference $C^{\mathrm{ref}}_{z,N}$, define

$$F_N(x) = (C^{\mathrm{ref}}_{z,N})^{-1/2} J_N(x) \Sigma_N^{1/2}, \qquad K_{E,N}(x) = F_N(x) F_N(x)^*. \tag{24}$$

The mean-map component has the exact identity

$$T_2^{(E)}(N;x) = \|K_{E,N}(x)\|^2_{\mathcal{S}_2} = \sum_j s_j(F_N(x))^4 = \|F_N(x)\|^4_{\mathcal{S}_4}. \tag{25}$$

For a stochastic VAE, let $S_{\mathrm{post},N}(x)$ be its conditional latent posterior covariance and take $C^{\mathrm{ref}}_{z,N}$ to be the clean encoder-mean covariance. Under the local, independent, zero-mean input perturbation model, write $W_N = (C^{\mathrm{ref}}_{z,N})^{-1/2}$. The local and population injected covariances are

$$\begin{aligned} K^{\mathrm{full}}_{E,N}(x) &:= K_{E,N}(x) + W_N S_{\mathrm{post},N}(x) W_N^*, \\ \overline{K}^{\mathrm{full}}_{E,N} &:= \mathbb{E}_x K^{\mathrm{full}}_{E,N}(x), \qquad T_2^{(E,\mathrm{full})}(N) := \left\| \overline{K}^{\mathrm{full}}_{E,N} \right\|^2_{\mathcal{S}_2}. \end{aligned} \tag{26}$$

The pointwise $K^{\mathrm{full}}_{E,N}(x)$ defines a local covariance-matched Gaussian proxy certificate. Its population average is generally a non-Gaussian mixture: $T_2^{(E,\mathrm{full})}$ is then a covariance budget, not a log-density tail certificate. We report it separately from the mean-map-only $T_2^{(E)}(N;x)$.

The implementable adapter is sample independent. We freeze one nested Fourier–channel basis across resolutions, apply the task-aware KKT rule to the diagonal structured component of $\overline{K}^{\mathrm{full}}_{E,N}$, and use the square-root amplitude gains in Equation (19). Because the implicit full operator need not commute with this basis, a final radial rescaling certifies

$$\widetilde{K}^{\mathrm{full}}_{E,N} = G_{\psi,N} \overline{K}^{\mathrm{full}}_{E,N} G^*_{\psi,N}, \qquad \left\| \widetilde{K}^{\mathrm{full}}_{E,N} \right\|^2_{\mathcal{S}_2} \le \tau_E \quad \text{for every audited } N. \tag{27}$$

The guarantee is for the declared population covariance, not every input pointwise. The gate changes only stochastic residual amplitudes in the fixed whitening geometry. An invertible adapter that transforms both representation and reference would instead preserve the singular values of F_N (Section G).

We audit the encoder pushforward separately from the downstream *marginal* diffusion diagnostic

$$K^{\mathrm{marg}}_{\mathrm{diff},N,t} = (C^{\mathrm{ref}}_{z,N})^{-1/2} (C_{z,N,t} - C^{\mathrm{ref}}_{z,N}) (C^{\mathrm{ref}}_{z,N})^{-1/2}. \tag{28}$$

This object compares time-t marginal covariances; it is not the conditional transition update used in $\mathcal{B}_{\mathrm{path}}$. Structured operators are summed exactly. For an implicit real operator, fixed independent Rademacher probes (Hutchinson, 1989) give

$$\widehat{T_2}_m(N) = \frac{1}{m} \sum_{r=1}^m \|K_N v_r\|_2^2, \qquad \mathbb{E}\widehat{T_2}_m(N) = T_2(N). \tag{29}$$

Training uses a separate cheap surrogate; final probes are independent of training, fixed across paired comparisons, and reported with operator calls, time, and memory. Probe variance, stopping rules, complex operators, and the difference between global certificates and blockwise diagnostics are detailed in Section H.

4 Experiments

We test one claim: at matched task utility, a finite raw-T_2 budget allocated by task sensitivity controls refinement and diffusion-time accumulation more effectively than scalar, frequency-only, or operator-norm controls. A controlled suite isolates the mechanism; one pinned latent-diffusion system tests whether the same criterion yields a useful quality–stability–cost frontier.

Shared protocol. Each sample is formed once at the highest resolution and antialiased to lower cutoffs. Every conditional Gaussian comparison reports raw T_2, $M = \|K\|_{\mathrm{op}}$, actual and certified positivity margins, and a per-operator tail radius at $\eta_{\mathrm{tail}} = 0.025$; an implicit audit reserves another 0.025 for estimation error. Population mixtures and non-Gaussian endpoints are labeled as covariance diagnostics. Structured sums are exact and implicit comparisons use one frozen final probe bank. Matched training, validation selection, shell diagnostics, and compute accounting are detailed in Section J.

4.1 Controlled mixed spectra and a high-frequency task

This suite is designed to falsify the explanation that Schatten budgeting is just low-pass filtering. It combines alternating dyadic decay, sparse spectral spikes, growing multiplicity, and the frequency-matched construction in Section I. The latter holds its Fourier power summary, trace, and operator norm fixed while changing raw T_2; exact cumulative mass supplies ground truth.

A conditional Gaussian generator carries its class label through a fixed sparse set of high-frequency covariance modes; low modes are label-independent nuisance. A classifier trained on samples from the unmodified target generator is then frozen. Gates act on the generator covariance before sampling, and the nonnegative q_j are squared sensitivities of conditional loss to covariance-mode deviations, estimated on the training split. Comparisons include no intervention, scalar rescaling, hard and smooth physical low-pass gates, an operator-norm cap, and the unweighted radial Schatten projection. Every method sweeps the same locked τ grid. Here p is one fixed isotropic Gaussian reference shared by all families and q is the corresponding gated conditional law. Conditional negative log likelihood is the primary task metric; label accuracy, raw T_2, and the exact fluctuation scale $\sqrt{T_2/2}$ of the centered $\log(dq/dp)$ under $X \sim q$ are secondary axes.

The first result figure will report shell mass, cumulative raw T_2 under refinement, and the complete task-quality–stability frontier. Its decisive comparison will test whether task-aware allocation retains the high-frequency label at a budget where frequency-only controls remove it.

4.2 Latent diffusion over resolution and time

We use one revision-pinned latent-diffusion pipeline and one fixed image–caption manifest with disjoint adapter-training, validation, audit, and test splits. The resolution ladder is $R \in \{256, 512, 1024\}$; nested image and noise realizations share their resolved physical modes.

Encoder intervention. We audit the population full stochastic VAE covariance $T_2^{(E,\mathrm{full})}$ from Equation (26), including posterior covariance. We separately report pointwise local Gaussian-proxy certificates for $K^{\mathrm{full}}_{E,R}(x)$ and the mean-map diagnostic $T_2^{(E)}(R;x) = \|F_R(x)\|^4_{\mathcal{S}_4}$. The proposed adapter is one sample-independent gate in a nested Fourier–channel basis, with a constraint on the finite adapter-training population at every resolution. The reference, basis, empirical $\overline{K}^{\mathrm{full}}_{E,R}$, and final radial factor are fit on that split and frozen; held-out audits never refit or rescale the gate. The real-suite q_j are training-split squared derivatives of a normalized sum of reconstruction LPIPS and the frozen denoising objective, averaged across the predeclared resolution–time sampling law and symmetry blocks, then frozen. Baselines are scalar rescaling, a smooth frequency gate, spectral normalization, a matched Jacobian/Frobenius penalty, and the unweighted Schatten projection. Frozen and adapted protocols change only covariance-gate parameters; the encoder, decoder, denoiser, and whitening reference remain fixed.

Reconstruction LPIPS (Zhang et al., 2018) is primary, with PSNR, reconstruction FID (Heusel et al., 2017), and texture-subset patch LPIPS as guardrails. The second figure will overlay frozen and adapted Pareto frontiers. The main table will report full-population raw T_2 and growth, pointwise certificate and mean-map summaries, quality, audit time, backward equivalents, and peak memory.

Diffusion-time accumulation. Let $p_{0:T}$ be one fixed positive-definite reference schedule and $q^g_{0:T}$ the law for gate g, obtained by adding its designed PSD covariance to that reference; the unfiltered target is one arm rather than the reference. All arms use $q^g_0 = p_0$ and the same conditional mean maps. Using the same checkpoint, data, and validation-selected operating points, we compute the conditional update $K^g_{R,t}(X_{t-1})$ between $q^g_t(\cdot \mid X_{t-1})$ and $p_t(\cdot \mid X_{t-1})$ for $T \in \{10, 50, 100, 250, 1000\}$. These conditionals define $\mathcal{B}_{\max}$ and $\mathcal{B}_{\mathrm{path}}$ for each arm; the weighted training diagnostic remains separate. For the nonlinear denoiser, paired path samples estimate $C^{p}_{R,T}$ and $C^{q^g}_{R,T}$ separately to obtain $\mathcal{B}_{\mathrm{end}}$; analytic covariance recursion is reserved for linear-Gaussian calibration. The marginal $K^{\mathrm{marg}}_{\mathrm{diff},R,t}$ is never summed as path risk. If an ablation changes the initial law or conditional means, its initial KL and mean-shift terms are reported separately. For a non-Gaussian endpoint, $\mathcal{B}_{\mathrm{end}}$ is reported as a covariance diagnostic. Generation FID (Heusel et al., 2017) is primary; semantic alignment and high-band error on a locked texture-prompt subset are guardrails. The third figure will show the resolution–time budget map together with quality at selected operating points.

5 Related Work

Noise and spectral control. Dropout uses stochastic feature perturbations (Srivastava et al., 2014), while spectral normalization bounds the largest singular value of a learned map (Miyato et al., 2018). Diffusion work has adjusted schedules for high resolution (Hoogeboom et al., 2023), learned multivariate noise (Sahoo et al., 2024), reweighted timesteps by signal-to-noise ratio (Hang et al., 2023), and designed schedules from frequency response (Benita et al., 2025). These approaches change complementary parts of the pipeline. Our constraint instead targets cumulative squared mass of a *whitened relative covariance* under resolution and time, and its task-aware allocation need not favor low frequency.

Latent and function-space diffusion. Latent diffusion (Rombach et al., 2022) applies Gaussian updates in an autoencoded representation (Kingma & Welling, 2014), reducing the cost of pixel-space diffusion (Ho et al., 2020). We treat the encoder's spectral effect as measurable rather than automatic: low finite dimension and good reconstruction do not imply uniform control along refinement. Function-space diffusion and neural operators study resolution-compatible laws (Kerrigan et al., 2023; Seidman et al., 2023), while regularized function-space objectives address well-posedness (Cinquin & Bamler, 2025). Gaussian equivalence and the Hilbert–Carleman determinant identify Hilbert–Schmidt relative updates as the asymptotic domain (Bogachev, 1998; Minh, 2022); our focus is the finite budget and the task decision inside that domain.

6 Conclusion

Schatten-2 control tells us which Gaussian perturbation families remain in the asymptotic feasible region; it does not by itself choose a useful operating point. Task-aware Schatten budgeting turns that boundary into a decision rule for latent diffusion. Raw finite-resolution mass and separate maximum, path, and end-state budgets quantify how covariance risk accumulates over resolution and time, while task-weighted allocation reserves more of the available budget for modes assigned high task sensitivity. The resulting perspective is neither scalar noise tuning nor generic low-pass filtering: it treats stability and task utility as an explicit Pareto decision under a measurable spectral budget. Structured accounting and scoped matrix-free audits give that decision an explicit computational contract.

AI use statement

Generative-AI tools assisted with outlining, prose drafting, code scaffolding, and consistency checks. The authors inspect the resulting text and code, independently verify mathematical and empirical claims, and retain full responsibility for the submitted work.

Ethics statement

This work develops mathematical diagnostics and learning algorithms and does not involve human participants, private data, or deployment decisions. We do not identify material ethical risks beyond the ordinary resource and misuse considerations associated with machine-learning research.

Reproducibility statement

The experiment protocol fixes two suites, three result figures, one result table, the same-source resolution ladder, primary and texture metrics, budget grid, go/no-go rules, and final audit probes before the main runs. For every reported result, the artifact will record the exact model revision, data-split manifest, quadrature convention, covariance reference, probe bank, training seed, raw output, figure-generation command, wall-clock time, backward equivalents, and peak memory. Final audit probes are never used for training or operating-point selection.

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A Carleman–Fredholm Finite Part

For a finite self-adjoint matrix K with $I + K \succ 0$, define

$$\det_2(I + K) := \det(I + K)e^{-\operatorname{Tr} K}. \tag{30}$$

The finite covariance objective in Equation (11) is exactly

$$\mathcal{L}^{\text{ren}}(K) = -\frac{1}{2}\log\det_2(I + K) = \frac{1}{2}\sum_j \big[\lambda_j - \log(1 + \lambda_j)\big]. \tag{31}$$

This identity removes the first-order trace term; it does not alter any finite-dimensional probability law. Its role becomes substantive under refinement. If $K \in \mathcal{S}_2$ and $I + K \succeq \delta I$, the canonical product

$$\det_2(I + K) = \prod_j (1 + \lambda_j)e^{-\lambda_j} \tag{32}$$

converges even when $K \notin \mathcal{S}_1$. Thus the ordinary trace and log-determinant need not exist separately, whereas their finite part does. If $K \in \mathcal{S}_1$, Equation (31) agrees with their usual difference.

When $\|K\|_{\text{op}} < 1$, the logarithm has the absolutely convergent expansion

$$-\log\det_2(I + K) = \sum_{k=2}^{\infty} \frac{(-1)^k}{k}\operatorname{Tr}(K^k). \tag{33}$$

If $\|K\|_{\text{op}} \le \rho < 1$, truncation after order $L \ge 2$ obeys

$$\left| -\log\det_2(I + K) - \sum_{k=2}^{L} \frac{(-1)^k}{k}\operatorname{Tr}(K^k) \right| \le \frac{\rho^{L-1}}{(L + 1)(1 - \rho)} \|K\|_{\mathcal{S}_2}^2. \tag{34}$$

Indeed, $|\operatorname{Tr}(K^k)| \le \rho^{k-2}\|K\|_{\mathcal{S}_2}^2$ and summing the geometric tail proves the claim. The main text uses the raw certificate $\big(\|K\|_{\mathcal{S}_2}^2, \|K\|_{\text{op}}, \min\{1, \lambda_{\min}(I+K)\}\big)$ and reports the unclipped geometric margin separately; it does not require numerically evaluating Equation (33).

B Proofs for the Finite Certificate

Let $g(\lambda) = \lambda - \log(1 + \lambda)$. Taylor's theorem in integral form gives, for every $\lambda > -1$,

$$g(\lambda) = \lambda^2 \int_0^1 \frac{1 - s}{(1 + s\lambda)^2}\, ds. \tag{35}$$

Under the assumptions of Theorem 3.1, every eigenvalue lies in $[-1 + \delta, M]$, so

$$\frac{\lambda^2}{2(1 + M)^2} \le g(\lambda) \le \frac{\lambda^2}{2\delta^2}. \tag{36}$$

Summing and multiplying by $1/2$ proves Equation (12). If $K \succeq 0$, then

$$g(\lambda) = \int_0^{\lambda} \frac{t}{1 + t}\, dt, \qquad \frac{\lambda^2}{2(1 + M)} \le g(\lambda) \le \frac{\lambda^2}{2}, \tag{37}$$

which gives the sharpened constants.

We next prove the exact fluctuation and its high-probability version. Work in whitened coordinates and diagonalize K. For $X = (I+K)^{1/2}\xi$ with $\xi \sim \mathcal{N}(0, I)$,

$$Y := \log \frac{dq}{dp}(X) - \mathbb{E}_q \log \frac{dq}{dp}(X) = \frac{1}{2}\sum_j \lambda_j(\xi_j^2 - 1). \tag{38}$$

Independence and $\mathrm{Var}(\xi_j^2) = 2$ give $\mathrm{Var}(Y) = \|K\|_{\mathcal{S}_2}^2/2$. Moreover, for $|t| < 1/M$,

$$\log \mathbb{E}e^{tY} = \frac{1}{2}\sum_j \big[-\log(1 - t\lambda_j) - t\lambda_j\big] \tag{39}$$

$$\leq \frac{t^2\,\|K\|_{\mathcal{S}_2}^2}{4(1-|t|M)}. \tag{40}$$

The inequality follows by expanding each term and using $|\lambda_j|^k \leq M^{k-2}\lambda_j^2$ for $k \geq 2$. Applying the same bound to Y and $-Y$, followed by the Chernoff bound for a sub-gamma variable with variance factor $\|K\|_{\mathcal{S}_2}^2/2$ and scale M, yields

$$\Pr\left(|Y| \geq \sqrt{\|K\|_{\mathcal{S}_2}^2\, x} + Mx\right) \leq 2e^{-x}. \tag{41}$$

Substituting $x = \log(2/\eta)$ proves the explicit radius

$$|Y| \leq \sqrt{\|K\|_{\mathcal{S}_2}^2 \log\frac{2}{\eta}} + M\log\frac{2}{\eta} \qquad \text{with probability at least } 1-\eta. \tag{42}$$

The equivalences in Theorem 3.2 now follow immediately from the two-sided objective bound and the exact variance identity, with the constants uniform in N. The divergence statement follows from the corresponding lower bounds. Finally, Theorem C.1 is the integral-test asymptotic for $\sum_{j \leq n_N} j^{-2\alpha}$. These results establish boundedness under refinement; they do not identify a continuum limit without compatibility.

C Compatible Continuum Realization

Let $\mathcal{H}$ be a separable real Hilbert space, let C be positive, injective, and trace class, and let $K = K^* \in \mathcal{S}_2$ satisfy $I + K \succeq \delta I$. Since K is bounded, $C^{1/2}(I+K)C^{1/2}$ is trace class and defines a Gaussian covariance. The uniform positivity gap makes its Cameron–Martin norm equivalent to that of C, while the relative covariance update is Hilbert–Schmidt; the two Gaussian measures are therefore equivalent.

Suppose compatible finite-rank lifts satisfy $K_N \to K$ in $\mathcal{S}_2$ and share the positivity gap. Continuity of $\det_2$ and an isonormal coupling give

$$\mathcal{L}_N^{\mathrm{ren}} \to -\frac{1}{2}\log\det\nolimits_2(I+K), \qquad \mathbb{E}|Y_N - Y|^2 = \frac{1}{2}\,\|K_N - K\|_{\mathcal{S}_2}^2 \to 0. \tag{43}$$

Uniform boundedness of $\|K_N\|_{\mathcal{S}_2}$ without $\mathcal{S}_2$ convergence gives the no-divergence statement of Theorem 3.2, but it does not assert convergence of an incompatible cutoff sequence.

Corollary C.1 (Power-law phase transition). *If $s_j(K) \asymp j^{-\alpha}$ and K_N retains the first n_N singular directions under the uniform conditions of Theorem 3.2, then*

$$T_2(N) \asymp \begin{cases} 1, & \alpha > \frac{1}{2}, \\ \log n_N, & \alpha = \frac{1}{2}, \\ n_N^{1-2\alpha}, & \alpha < \frac{1}{2}. \end{cases} \tag{44}$$

This is an integral-test asymptotic; finite audits still report $T_2(N_{\max})$ and the independent variable used for any fitted slope.

D PRECISION–COVARIANCE BRIDGE

Suppose $H = H^*$, $\|H\|_{\mathrm{op}} \le \rho < 1$, and $K = H(I-H)^{-1}$. Functional calculus gives

$$\frac{1}{1+\rho}\,\|H\|_{\mathcal{S}_2} \le \|K\|_{\mathcal{S}_2} \le \frac{1}{1-\rho}\,\|H\|_{\mathcal{S}_2}\,. \tag{45}$$

Thus the two parameterizations have the same dimension-uniform $\mathcal{S}_2$ boundary. If $p = \mathcal{N}(0, C)$ and $q = \mathcal{N}(0, C^{1/2}(I-H)^{-1}C^{1/2})$, then

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\left[\operatorname{Tr} K + \log\det(I-H)\right] = -\frac{1}{2}\log\det{}_2(I+K), \tag{46}$$

$$D_{\mathrm{KL}}(p\|q) = -\frac{1}{2}\left[\operatorname{Tr} H + \log\det(I-H)\right] = -\frac{1}{2}\log\det{}_2(I-H). \tag{47}$$

For $Z \sim p$ in whitened coordinates,

$$\log\frac{dq}{dp}(Z) - \mathbb{E}_p\log\frac{dq}{dp}(Z) = \frac{1}{2}\left(\langle Z, HZ\rangle - \operatorname{Tr} H\right), \qquad \operatorname{Var}_p\left[\log\frac{dq}{dp}(Z)\right] = \frac{1}{2}\,\|H\|^2_{\mathcal{S}_2}\,. \tag{48}$$

Under q, the centered quadratic is instead governed by K and has variance $\|K\|^2_{\mathcal{S}_2}/2$. This specifies both the sampling measure and the KL orientation for precision-space implementations.

E TASK-AWARE PROJECTION

The program in Equation (17) is a convex quadratic program. With multiplier $\mu/2 \ge 0$ for the squared Hilbert–Schmidt budget, its separable Lagrangian (before imposing the box) is

$$\frac{1}{2}\sum_j q_j(\widetilde{\lambda}_j - \lambda_j)^2 + \frac{\mu}{2}\left(\sum_j \widetilde{\lambda}_j^2 - \tau\right). \tag{49}$$

Stationarity gives $(q_j + \mu)\widetilde{\lambda}_j = q_j\lambda_j$. Since $q_j, \lambda_j \ge 0$, projection onto $[0, \rho_{\mathrm{cap}}]$ therefore yields

$$\widetilde{\lambda}_j(\mu) = \min\left\{\rho_{\mathrm{cap}}, \frac{q_j}{q_j+\mu}\lambda_j\right\}. \tag{50}$$

At $q_j = \mu = 0$ the objective is flat in that coordinate; choosing zero gives a deterministic minimum-budget solution. If the $\mu = 0$ solution is feasible, complementary slackness selects it. Otherwise $S(\mu) = \sum_j \widetilde{\lambda}_j(\mu)^2$ is continuous and decreases from a value above τ to zero, so one-dimensional bisection finds the unique budget-active solution (with the $\tau = 0$ case understood as the limit $\mu \to \infty$). This derivation relies on the fixed commuting basis and a positive-semidefinite quadratic task metric; it does not apply unchanged to a learned basis or to an indefinite first-order score.

For a fixed compatible basis, a nonnegative envelope w_j satisfying $\sum_j w_j^2 \le 1$ gives the task-agnostic modewise alternative

$$\phi_{\tau,j}(\lambda) = \min\{\lambda, \sqrt{\tau}\,w_j\}, \qquad \lambda \ge 0, \tag{51}$$

which also enforces squared Hilbert–Schmidt mass at most τ. After any covariance gate, sampling uses a factor $B^{(\tau)}(B^{(\tau)})^* = K^{(\tau)}$:

$$\epsilon \sim \mathcal{N}(0, I) \text{ independently of } z, \qquad \xi^{(\tau)} = C^{1/2}B^{(\tau)}\epsilon, \qquad z' = z + \xi^{(\tau)}. \tag{52}$$

F TEMPORAL BUDGET IDENTITIES

For Markov laws with the required absolute continuity, factorization of their joint densities and iterated expectation give

$$\log\frac{dq_{0:T}}{dp_{0:T}} = \log\frac{dq_0}{dp_0} + \sum_{t=1}^{T}\log\frac{dq_t(\,\cdot \mid X_{t-1})}{dp_t(\,\cdot \mid X_{t-1})}, \tag{53}$$

which proves Equation (53). Conditional on $\mathcal{F}_{t-1}$, equal-mean Gaussian transitions have covariance KL $\mathcal{L}^{\mathrm{ren}}(K_t)$. Applying Theorem 3.1 conditionally and then taking expectation proves the two-sided path-KL bound by fixed multiples of $\mathcal{B}_{\mathrm{path}}$. If the transition means differ, the additional term is

$$\frac{1}{2}(m_t^q - m_t^p)^*(C_t^p)^{-1}(m_t^q - m_t^p), \tag{54}$$

and is outside the covariance certificate.

Let ℓ_t denote the conditional covariance log-ratio and

$$d_t = \ell_t - \mathbb{E}_q[\ell_t \mid \mathcal{F}_{t-1}]. \tag{55}$$

Then (d_t) is a martingale-difference sequence. Conditional application of Equation (38) yields

$$\mathbb{E}_q[d_t^2 \mid \mathcal{F}_{t-1}] = \frac{1}{2}T_2(K_t). \tag{56}$$

Orthogonality of martingale differences proves

$$\mathrm{Var}_q\left(\sum_{t=1}^{T} d_t\right) = \frac{1}{2}\sum_{t=1}^{T}\mathbb{E}_q T_2(K_t) = \frac{1}{2}\mathcal{B}_{\mathrm{path}}. \tag{57}$$

For deterministic weights, the corresponding variance is $\frac{1}{2}\sum_t w_t^2 \mathbb{E}_q T_2(K_t)$. This square on w_t is distinct from the linear weighting in a training loss. Raw increments ℓ_t are not in general martingale differences, and state-dependent or data-dependent weights require the corresponding predictable-weight formulation.

The endpoint budget is different again. For a linear Gaussian recursion $X_t = A_t X_{t-1} + \epsilon_t$ with innovation covariance Q_t, for example,

$$C_t = A_t C_{t-1} A_t^* + Q_t. \tag{58}$$

One must run this recursion under both laws and whiten their final covariance difference as in Equation (23). Neither the path chain rule nor the martingale variance identity turns this endpoint covariance into a sum or product of stepwise relative updates.

G Encoder Audit and Coordinate Invariance

For F of finite rank with singular values $s_j(F)$,

$$\|FF^*\|_{\mathcal{S}_2}^2 = \mathrm{Tr}\big((FF^*)^2\big) = \mathrm{Tr}\big((F^*F)^2\big) = \sum_j s_j(F)^4. \tag{59}$$

This proves Equation (25). It also explains why an operator-norm cap alone is insufficient: if $r = \mathrm{rank}(F)$, then

$$\|F\|_{\mathcal{S}_4}^4 \le r\,\|F\|_{\mathrm{op}}^4\,, \tag{60}$$

so growing effective rank can consume the squared Hilbert–Schmidt budget even when the largest singular value is controlled.

For an invertible raw-coordinate adapter A, let

$$C' = ACA^*, \qquad F' = (C')^{-1/2}AJ\Sigma^{1/2} = QF, \qquad Q = (C')^{-1/2}AC^{1/2}. \tag{61}$$

Direct multiplication gives

$$QQ^* = (C')^{-1/2}ACA^*(C')^{-1/2} = I. \tag{62}$$

In equal finite dimensions with invertible A, Q is unitary; hence QF and F have identical singular values. A genuine repair must therefore modify the noise covariance in one declared whitening geometry, as in Equation (27), or change the represented support. If the reference is re-estimated after fitting the gate, the intervention and audit geometry change together and the certificate must be recomputed.

H MATRIX-FREE ESTIMATORS AND PROBE COST

Let $A = K^*K$. For an unnormalized real Rademacher probe v, $\|Kv\|^2 = v^\top A v$ and

$$\mathbb{E}[v^\top A v] = \operatorname{Tr} A = \|K\|_{\mathcal{S}_2}^2, \qquad \operatorname{Var}(v^\top A v) = 2\sum_{i\neq j} A_{ij}^2. \tag{63}$$

Averaging m independent probes divides this variance by m. Consequently probe count is governed by off-diagonal mass in the probe basis and the desired confidence width, not directly by model parameter count. For complex Fourier operators, one may either use independent unit-circle probes with $\mathbb{E}[vv^*] = I$ or realify the operator; the probe law and multiplicities must be recorded.

Structured diagonal and low-rank updates should be summed exactly. For an implicit encoder update $K = FF^*$, one application of Kv requires one application each of F^* and F, which corresponds to a VJP and a JVP together with the whitening and input-covariance maps. Differentiating this quantity through encoder parameters may require second-order autodiff. A truncated regularized determinant estimator uses $O(mL)$ operator–vector products and has deterministic truncation error Equation (34); it is not required for the raw-T_2 audit.

An online one-probe value is a stochastic training surrogate, not a finite certificate. A final audit uses probes drawn independently of training and fixed across paired methods. Across resolutions, a common bank is valid only after specifying a coupling of compatible nested spaces. Optional stopping requires a sequentially valid confidence procedure or a predeclared probe batch schedule; repeatedly inspecting a fixed-m confidence interval does not preserve its nominal coverage. Common random numbers often improve a paired difference empirically, but this is not guaranteed without controlling the covariance of the two estimators. Finally, a blockwise or localized surrogate certifies only that block; it does not inherit the global theorem.

I FREQUENCY-MATCHED CONTROLS

For completeness, let $d = m^2$ and define two nonnegative spectra by

$$\lambda_j^{\text{stable}} = \begin{cases} \eta, & j = 1, \\ \eta/(m+1), & 2 \le j \le d, \end{cases} \qquad \lambda_j^{\text{unstable}} = \begin{cases} \eta, & 1 \le j \le m, \\ 0, & m < j \le d. \end{cases} \tag{64}$$

Let U_d be a fixed flat unitary and set $A_d^{(a)} = U_d \operatorname{diag}(\lambda^{(a)}) U_d^*$ for $a \in \{\text{stable}, \text{unstable}\}$. A flat unitary has constant-modulus entries, so each diagonal entry of $U_d \operatorname{diag}(\lambda) U_d^*$ equals $d^{-1}\sum_j \lambda_j$. Both spectra have trace ηm, giving the shared diagonal η/m. Their largest eigenvalue is η, while direct summation gives

$$\sum_j \left(\lambda_j^{\text{stable}}\right)^2 = \eta^2 \left[1 + \frac{m^2-1}{(m+1)^2}\right] = \eta^2 \frac{2m}{m+1}, \tag{65}$$

$$\sum_j \left(\lambda_j^{\text{unstable}}\right)^2 = m\eta^2. \tag{66}$$

Thus the stable family has bounded Hilbert–Schmidt mass but diverging trace, whereas the unstable family leaves $\mathcal{S}_2$.

To match spatial-frequency appearance, let

$$G_R = \operatorname{diag}\left[(1 + \|k\|^2)^{-s/2}\right]_{\|k\| \le R}, \qquad s > 1, \tag{67}$$

and set $K_{R,d}^{(\cdot)} = G_R \otimes A_d^{(\cdot)}$. The two families have the same Fourier diagonal and hence the same radial power summary. For every spatial mask M_R,

$$\left\| (M_R G_R M_R) \otimes A_d^{(\cdot)} \right\|_{\mathcal{S}_2}^2 = \|M_R G_R M_R\|_{\mathcal{S}_2}^2 \left\| A_d^{(\cdot)} \right\|_{\mathcal{S}_2}^2. \tag{68}$$

Any frequency-only control that retains a nonzero low-frequency response has $\|M_R G_R M_R\|_{\mathcal{S}_2} \ge c > 0$ and therefore preserves the unstable family's $\sqrt{d}$ divergence. The comparison matches Fourier diagonal, radial power, trace, operator norm, bandwidth, low-frequency response, parameter count, probe count, and compute budget. Closed-form spectra provide the ground truth; Hutchinson probes only validate the matrix-free estimator.

J Additional Experimental Details

Resolution and covariance contract. Each natural image and controlled sample is formed once at the highest resolution and restricted with one fixed antialiasing operator. Split membership, augmentations, mode labels, and random seeds are shared across the ladder. Adjoint operators use the quadrature-weighted inner product of their discretization. At every intervention, B_R denotes a noise amplitude and $K_R = B_R B_R^*$ the whitened relative covariance. We report $T_2(R) = \mathrm{Tr}(K_R^* K_R)$ without division by pixels, channels, latent dimension, or effective rank; a diagonal amplitude b_j therefore contributes b_j^4. Rank shells and physical-frequency shells are never conflated.

Final probes and paired comparisons. Diagonal and explicitly low-rank operators are summed exactly. For an implicit operator, a disjoint calibration split selects the final probe count from a predeclared grid using absolute and relative precision targets. The selected count, support regularization, whitening, and estimator hyperparameters are then frozen. A public seed manifest generates one audit bank; compatible nested spaces share probe signs on common modes, and every baseline, budget, seed, and time uses the same bank. Training uses a separate fresh one-probe stream and never observes final probes. We will validate the estimator against exact materialization at the smallest feasible cutoff and report probe uncertainty separately from image and adaptation-seed variation.

For an implicit Gaussian comparison, we set $\eta_{\mathrm{est}} = 0.025$ for the probe upper-confidence bound and $\eta_{\mathrm{tail}} = 0.025$ for the conditional log-density tail, giving total failure probability at most 0.05 by a union bound. Structured exact comparisons use the same η_{tail} and have no probe error. These probabilities are per audited operator. Any simultaneous statement over the full resolution–time–arm grid uses a predeclared Bonferroni allocation and displays the adjusted radii. Positive injected-covariance operators have actual margin at least one, hence $\delta^{\mathrm{cert}} = 1$; when no tighter certified cap is available, the tail radius uses $M \le \|K\|_{\mathcal{S}_2} = \sqrt{T_2(K)}$ with the upper confidence bound substituted. A signed operator receives a tail certificate only when a valid lower positivity bound is available; otherwise $(M, \delta^{\mathrm{act}})$ is explicitly labeled a numerical diagnostic.

All learned comparisons match data, update count, optimizer, adapter parameter count, and three adaptation seeds. Confidence intervals use paired examples and common random numbers. Primary quality, texture quality, raw T_2, and cost remain separate axes rather than being collapsed into a single score. The quality non-inferiority margins and maximum audit-cost multiplier are stored in the locked configuration before the main seeds. Every audit reports operator–vector products, backward equivalents, wall-clock time, and peak accelerator memory.

Controlled mechanism test. The controlled families include alternating dyadic blocks with decay exponents 0.4 and 0.6, a front-heavy spectrum with a light tail, a rapidly decaying tail with sparse spikes, and the multiplicity construction in Section I. Exact cumulative mass is the membership decision; no fitted exponent replaces it. Conditional samples are generated on the finest Fourier grid. The binary label changes the covariance on a fixed sparse set of high modes below the Nyquist frequency of every reported cutoff; low modes have the same nuisance law in both classes. A four-layer convolutional classifier with global pooling is trained on samples from the unmodified base-resolution generator and frozen. Squared sensitivities of conditional loss to covariance-eigenvalue deviations are estimated on the training split, averaged within symmetry blocks, and frozen. All covariance gates are converted back to amplitude factors before sampling. The same validation-locked τ grid is used for no intervention, scalar scaling, hard and smooth low-pass filters, an operator-norm cap, radial projection, and task-aware projection.

Pinned latent-diffusion system. The real suite pins a Stable Diffusion v1.x-compatible pipeline with the `stabilityai/sd-vae-ft-mse` encoder/decoder and its matching frozen denoiser. Exact model revisions, scheduler configuration, preprocessing, latent scaling, and data-manifest hashes are artifact inputs. The full stochastic VAE pushforward includes both the encoder-mean Jacobian term and posterior covariance from the law of total covariance. The clean encoder-mean covariance is the latent reference and is estimated on the training split. One sample-independent Fourier–channel basis is fixed at the largest resolution and restricted to lower cutoffs. The task-aware KKT rule acts on the population operator's diagonal structured component; a final radial factor enforces the full-population budget on the finite adapter-training manifest at every R. The empirical $\overline{K}_{E,R}^{\mathrm{full}}$, basis, reference, and radial factor are then frozen. Held-out audit and test manifests estimate their own population covariance only for evaluation; they never update or rescale the gate. A held-out budget violation is reported as such. Per-image Schatten-4 values remain a separate distributional diagnostic.

For a finite empirical population $\overline{K} = n^{-1} \sum_i K(x_i)$, a population Hutchinson probe first forms $\overline{u} = n^{-1} \sum_i K(x_i)v$ and then records $\|\overline{u}\|^2$. Averaging $\|K(x_i)v\|^2$ over examples estimates a different pointwise quantity and is used only when that quantity is explicitly requested.

For the real suite, define

$$\mathcal{L}_{\text{task}} = \frac{\mathcal{L}_{\text{LPIPS}}}{\overline{\mathcal{L}}^0_{\text{LPIPS}}} + \frac{\mathcal{L}_{\text{denoise}}}{\overline{\mathcal{L}}^0_{\text{denoise}}}, \tag{69}$$

where the denominators are unfiltered means on the adapter-training split. For every mode, q_j is the training-split average of the squared derivative of Equation (69) with respect to its covariance eigenvalue. The average uses a predeclared uniform draw over reported resolutions and the pipeline's frozen training-time timestep law; a mode is averaged only over resolutions where it exists. Values are averaged within conjugacy and channel symmetry blocks and frozen before validation. Texture metrics are evaluation guardrails and do not enter q_j.

In the adapted protocol only the covariance gate is updated; the encoder, decoder, denoiser, basis, and reference remain fixed, so all arms stay in the whitening geometry of Equation (27). Any encoder-unfreezing ablation must re-estimate its reference and is not used for the main claim.

Per-image $K^{\text{full}}_{E,R}(x)$ values receive only the local covariance-matched Gaussian proxy certificate defined in the main text. The population mixture receives raw covariance diagnostics and no density-ratio tail claim.

The texture-rich reconstruction subset is selected before training using source-image high-band energy alone. The corresponding generation subset is selected from prompts by a predeclared rule independent of generated images. Operating points are selected on validation data as: best quality under a raw T_2 ceiling, best raw T_2 within the quality margin, and the frontier knee. No test metric participates in selection. For the temporal audit, one fixed positive reference schedule p is chosen before evaluation. Every gate defines q^g by adding its PSD innovation covariance to the same reference and shares $q^g_0 = p_0$ and the conditional mean map. Conditional $K^g_{R,t}$ defines maximum and path budgets. For the nonlinear denoiser, $C^p_{R,T}$ and $C^{q^g}_{R,T}$ are estimated from paired endpoint samples under the two full path laws, using the law of total covariance when within-sample stochasticity is present. Their whitened difference gives the endpoint covariance diagnostic. Analytic scheduler covariance recurrence is used only for linear-Gaussian calibration, including constant and idealized variance-preserving schedules.

Claim gates and artifact record. The controlled claim requires agreement between exact and stochastic raw T_2, separation of the frequency-matched pair, and a nondominated task-aware frontier segment that retains the high-frequency signal. The encoder claim requires a nondominated segment under both main and texture-quality margins. The temporal claim additionally requires paired $\mathcal{B}_{\text{path}}$ and $\mathcal{B}_{\text{end}}$ improvement at matched generation quality; a single-step reduction is insufficient. A cost claim additionally requires the locked time and memory caps. A failed gate removes that claim without changing the test protocol.

The final artifact will record all resolution ladders, splits, seeds, confidence intervals, hyperparameters, discretization weights, covariance references, probe banks, scheduler recurrences, raw outputs, figure commands, and compute measurements.